Statistical Analysis On Typing Speed and Accuracy Across Different Passage Types and Keyboard Types

Michael Dennis
Jack Kroll

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Introduction

This project looked to investigate various factors that could potentially influence how fast a user could type, and with what error rate. This was tested through two methods, altering the user interface to the computer, and the passage type that the user is typing. These findings are likely only relevant for those of similar age, college age, and those who regularly utilize both a keyboard and mobile device. Both of those descriptors keep consistent the typists understanding of language and what content is commonly used, and that they are also reasonably proficient in interfacing with a computer.

The first consideration was changing how a user could interface with the computer, we first selected a keyboard with thicker keys that has a greater tactile feel to it and requires more force to press down. Secondly, we selected a lower profile keyboard with much thinner keys that require far less force to press down and provide much less tactile feel. Finally, we selected a mobile keyboard with no physical buttons. The second variable we considered was the actual content that the user was going to be typing. We first selected news article content that wasn't technical (science, technology, etc) and was unlikely to be challenging to the typist. Secondly, we had chosen children's book content for a passage type, often with smaller and more similar words. Finally, we had selected Old English content taken from selected Shakespeare works that contained words a typist may not regularly use and that may not follow the same grammatical tendencies a typist is used to.

The investigation of these could potentially help to determine what keyboards provide generally faster typing speeds, and/or a lower error rate to the user. The altering of passage types could provide insight into if there exists some form of muscle memory that Old English could disrupt. Furthermore, a difference between kids books and news articles could indicate a relationship between typing speed and the general difficulty of words through length/syllable count. Both of these variables seek to investigate how either a user is disrupted physiologically through keyboard interface, or in some form psychologically through the kind of content that is being typed out.

Methods

This experiment was conducted as a controlled experiment as two separate variables were altered throughout the experiment. The data for this experiment was collected through a typing test administered online through monkeytype.com. Collection consisted of various testing sessions where several typing tests of varying conditions were conducted. Prior to each testing session, the participants were instructed to perform a "warm up" typing test using the standard typing test available on the website. This basic test consists of a 30 second typing test that contains a random assortment of random English words. This was included to ensure that each subsequent test within the testing session was performed under similar conditions and that the first test collected was not performed under significantly unique circumstances than the rest. In a similar manner, participants were encouraged to limit their given sessions to only a few tests per day to help prevent fatigue from setting in.

All tests were performed in the J. Robert Van Pelt and John and Ruanne Opie Library without any headphones/earbuds and were performed after 3PM on weekdays or on weekends to avoid peak hours which could potentially be louder and more disruptive to test takers. For our controlled variables, we selected 3 different keyboard types that were amply available in the testing environment, a “chunky” keyboard which was selected to be the “Lenovo Preferred Pro II USB Wired Keyboard”. Secondly, for the “low profile” keyboard we selected the “Dell Wired Keyboard - KB216” . Finally the mobile keyboard was the keyboard of the participants personal mobile device, to ensure that it was accessible to them and something they were comfortable using.

Each typing test consisted of 400 characters including spaces, which was a hard cutoff that occasionally resulted in truncated words and sentences. The news article content was taken from non-technical news articles published within a few weeks of November 1st 2025, from The New York Times, CNN, and Reuters. The children's books were taken from the website Monkeypen, and specifically from the two books “Ginger The Giraffe” and “SUNNY MEADOWS WOODLAND SCHOOL” which were both in the categories of ages 4-7 and ages 8-12. Finally, the Old English excerpts were taken from Shakespeare works. They were taken from the MIT distribution of the texts, and the prompts utilized were taken from various segments of “All's Well That Ends Well”. Due to the nature of these excerpts being from plays that included many breaks in the text, the collection specifically targeted longer segments of text that fulfilled the majority (if not all) of the 400 character limit.

Following the completion of the 400 character typing test, the words per minute was reported as the raw words per minute (WPM) as opposed to standard WPM reported by monkeytype.com which will adjust the reported WPM with the error rate you achieved to give a “practical WPM”. Instead, participants simply collected both metrics, raw WPM and error rate (%), to provide greater flexibility in the data analysis. Additionally, the exported data contains a metric regarding the percentage of the prompt that was not a standard A-Z (upper and lower case, and spaces) character which could also have an influence on the output WPM or error rate with key strokes requiring a further reach, or even simultaneous inputs which could make a more significant different in the participants WPM or error rate than the other variables previously considered.

Results

Stat	Typing Speed (WPM)	Accuracy (%)
Mean	72.86	94.44
Median	73	95
SD	8.52	2.02
Skew	-0.33	-0.44
IQR Range	69-78	93-96

Table 1: Summary Statistics

Table 1 shows the sample statistics for the entire survey, combining both typists. For typing speed, the mean was approximately 72.86 words per minute, which was pretty similar to the median, which was 73 words per minute. The data only had a mild skew of -0.33, although this shows that the data had a long tail, as shown in Figure 2 below. The standard deviation was 8.52 words per minute, with an IQR range between 68-78 WPM.

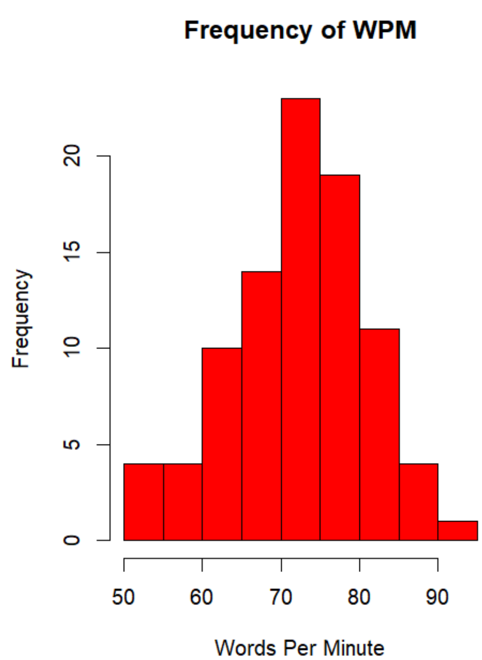

Figure 2 Frequency of Worder Per Minute Histogram

As for accuracy, the average score was 94.44% for the mean and 95% for the median. The standard deviation however, was significantly smaller, only being around 2.02% showing the typing accuracy had less deviation between tests compared to typing speed. The skew was -0.44, showing a moderate negative skew to the data with an IQR range between 93-96%.

Stat	Typing Speed (WPM)	Accuracy (%)
Mean	72.02	94.11
Median	72	94
SD	5.82	1.92
Skew	0.76	-0.80
IQR Range	69-75	93-95

Table 3: Subject A Summary Statistics

Stat	Typing Speed (WPM)	Accuracy (%)
Mean	73.69	94.78
Median	78	95
SD	10.57	2.09
Skew	-0.65	-0.26
IQR Range	66-81	93-96

Table 4: Subject B Summary Statistics

Before comparing the two subjects to each other, it is helpful to have summary statistics for both. There are a few striking similarities and differences between the two. As can be seen in Table 3 and Table 4, respectively, both users had a very similar mean typing speed, with user A averaging 72.02 WPM and user two averaging 73.69 WPM. Yet their medians are significantly different, with subject A having a median of 72 WPM and subject B having a median of 78 WPM. Subject B was also quite a bit less consistent than subject A, having an SD almost twice as large, which was represented by subject B also having a larger IQR range. Most interestingly, though, is how both have moderate to high skews but in completely different directions, with user B having a positive skew of 0.76 and user A having a negative skew of -0.65. This can be seen in the normal

QQ-plots of both typists' WPM seen in Figure 5 below, with both subjects having deviations from normality.

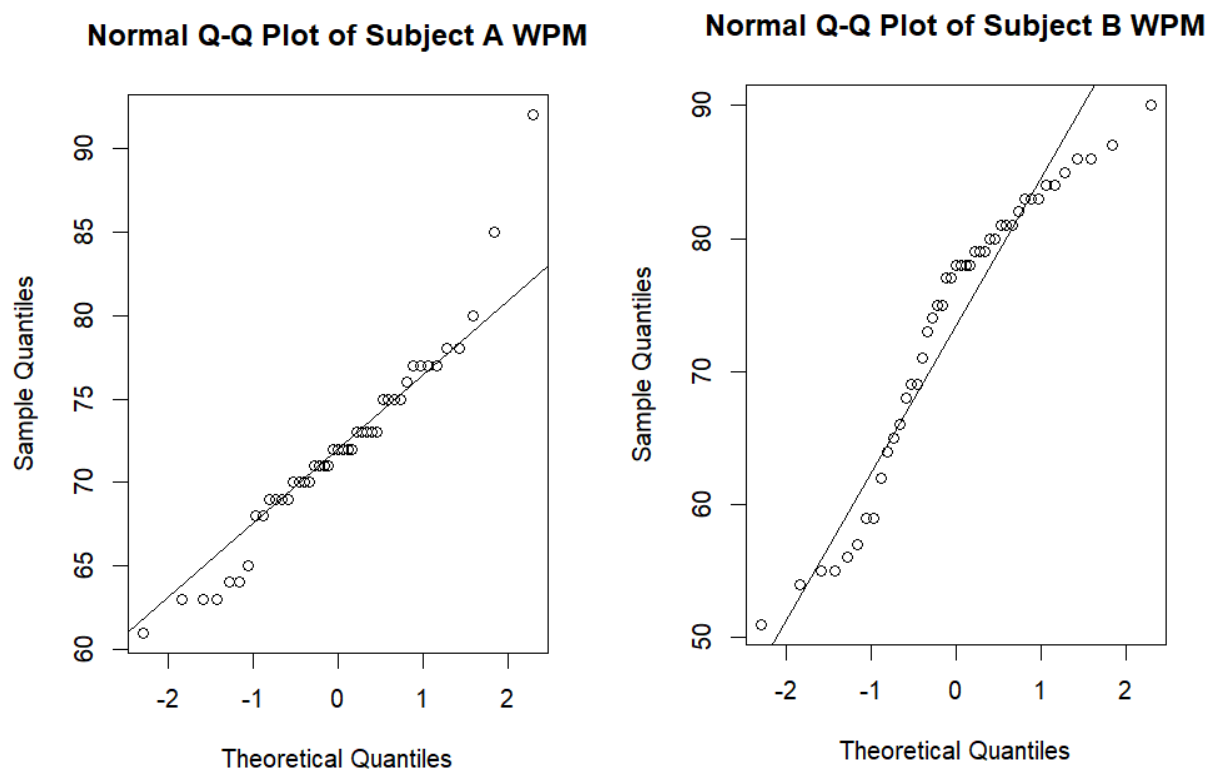

Figure 5: Normal Q-Q Plots of Subject A and B Typing Words Per Minute

This is a bit worrying due to the fact that T-tests have an assumption of normality, which is being broken by both groups. Yet due to the CLT theorem, with a significantly large n the t-test becomes robust to violations of the normality assumption. The experiment was balanced with both users having 45 trials, which makes the t-test robust to violations. A Welch's t-test will be used as the homoscedasticity seems to be false from Tables 3 and 4.

This T-test will be a two-sided Welch's T-test with a significance level of 0.9
H0: There is no difference in mean typing speed between users.
HA: There is a difference in the mean typing speed between users.

T-Value	P-Val	Conf-Interval (WPM)
-0.93	0.36	-4.67 - 1.33

Table 6: Welch's T-Test Results For Mean Typing Speed (WPM)

Through the results of the T-test, H0 will not be rejected in favor of the alternative. There is no statistically significant difference in average typing speed between the two typists. Yet due to the strong differences that can be seen from the SD, median, and skew that the typists do seem to have significant differences, even if the median of both sets is not one of them. Due to how significantly different the distributions are from each other, it is difficult to find a test without its assumptions being violated. Yet the Kolmogorov-Smirnov distribution was designed to test if two distributions are different. The assumptions are that the data is independent and continuous, although the Kolmogorov-Smirnov test is robust to continuous data violations with a small enough bin size, which is true here. Running the test gives Table 7 below.

H0: There is no difference in the distribution of typing speed between the two typists.
HA: There is a difference in the distribution of typing speed between the two typists.

D	P-Val
0.4	0.00079

Table 7: Kolmogorov-Smirnov Results for Typing Speed (WPM)

As can be seen in Table 7 above the distributions do have a statistically significantly different result, so HA is accepted over the Null hypothesis that there likely is a difference in the distributions of typing speed between the two typists.

The next analysis will be on the accuracy between the two subjects. Quickly looking at the data, it seems like the means are again not very different between the two typists on Figure 4 and Figure 5 they only differ by 0.67% which is inside the standard deviation for both typists. The medians are also relatively close, being 94% and 95% for typists A and B, respectively. The skew is a bit closer between the two subjects than WPM, although for typist A, there is a moderate skew of -0.80, and for typist B it is a mild skew of -0.26. The IQR ranges also overlap pretty heavily with typist A having an IQR between 93%-95% compared to typist B's 93%-96%. This is similarly shown on the box plot in Figure 8 below.

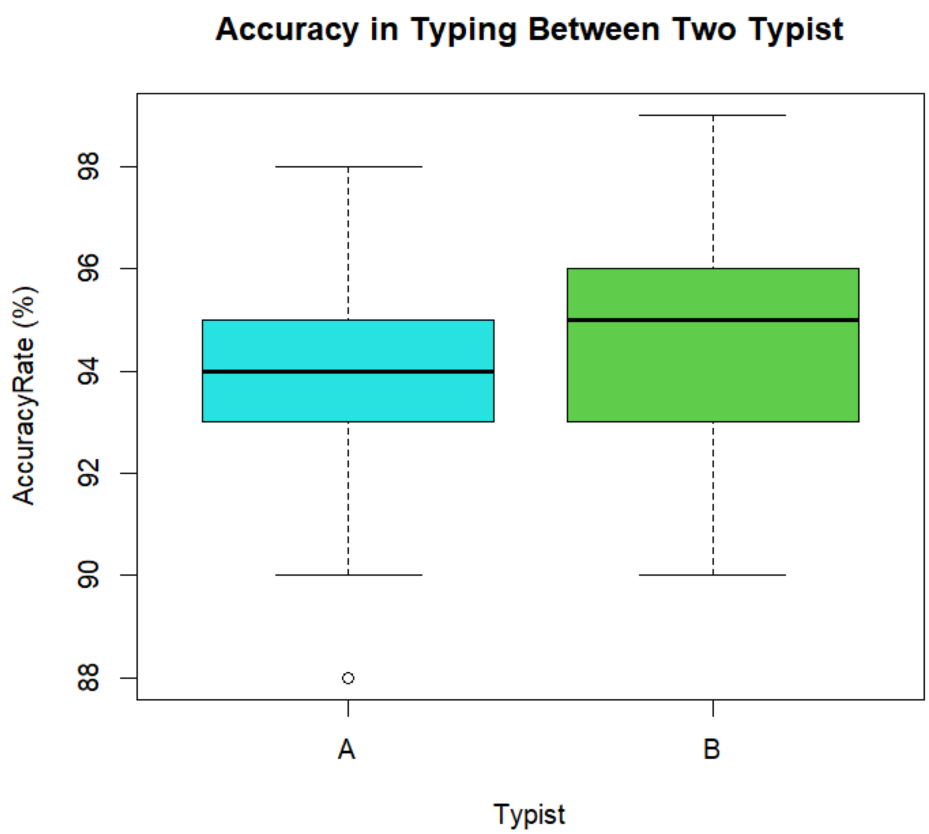

Table 8: Boxplot in Accuracy in Typing Between Two Typists

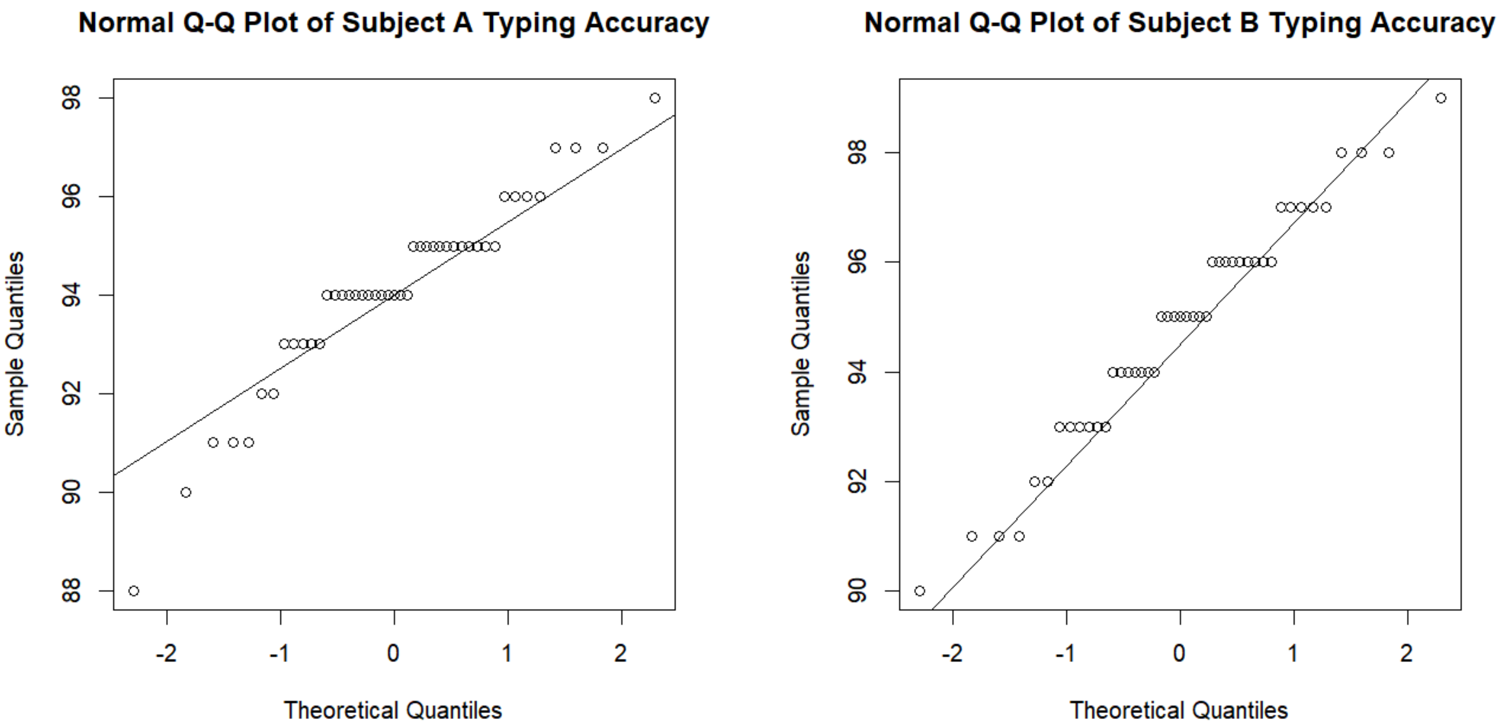

Figure 9: QQPlots of Typing Accuracy for both subjects

Before running a T-Test, it is a good idea to check the normality of the data which was done in Figure 9 below

Figure 9: QQPlots of Typing Accuracy for both subjects

The normality is quite a bit better here than it was for the Typing Speed. I feel good about the normality assumption for the T-Test.

For this T-Test again, homogeneity of variance will not be assumed and a two-sided Welch's T Test will be done with a significance level of 0.9

H0: There is no difference in mean typing accuracy between typists.
HA: There is a difference in the mean typing accuracy between typists.

T-Value	P-Val	Conf-Interval (Accuracy % Difference)
-1.58	0.12	-1.37 - 0.04

Figure 10 T-Test Results for Typing Accuracy Between Typists

While this is a borderline result and would warrant further research, H0 will not be rejected in favor of the alternative that there is a difference in the accuracy percentage between the two users.

While there were not any significant differences found between the two typists in terms of the mean, there is curiosity about how the three numerical variables (WPM, Accuracy, and Non-AZ Characters) in the passage correlate to each other. Figure 11 below shows the results of the correlation.

Numerical Variable	WPM	Accuracy (%)	NonAZChar (%)
WPM	1.00	0.48	-0.23
Accuracy (%)	0.48	1.00	-0.11
NonAZChar (%)	-0.23	-0.11	1.00

Figure 11: Correlation of Typing Statistics

Outside of the trivially high correlations to variables to themselves at a quick glance, there appears to be a high correlation between accuracy and WPM. Another correlation is how there seems to be a negative correlation between the percent non-AZ characters in passages with typing speed, this also makes sense due to how most typers are accustomed to typing on the AZ characters more often. The trends can also be seen in the scatter plot matrix in Figure 12

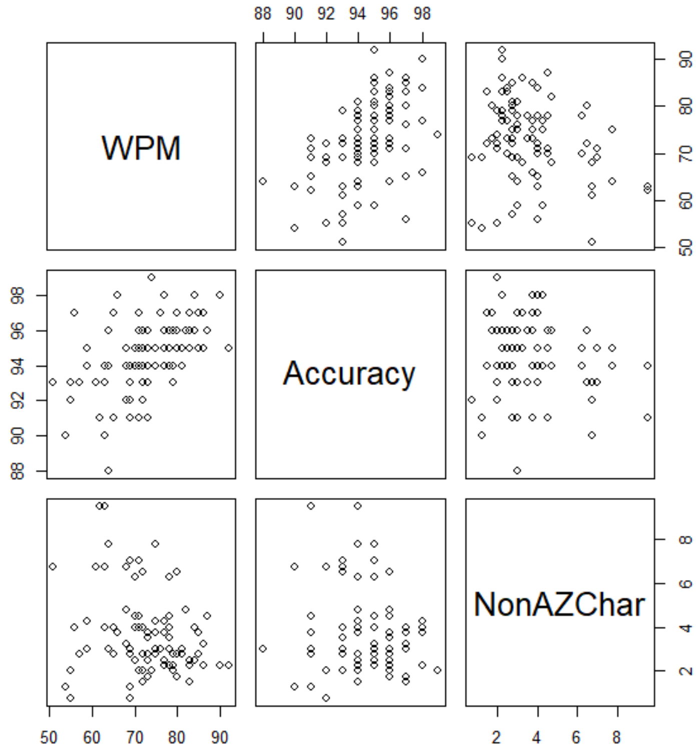

Figure 12: Scatter Plot Matrix of Typing Statistics

To be more thorough, a correlation test was found that only requires that the data has a positive or negative correlation and is independent, which is true for the data being tested, called the Spearman Test. For the first test we ran the Spearman Test on words per minute and accuracy with a significance level of 0.9

H0: There is no correlation between words per minute and accuracy
HA: There is a correlation between words per minute and accuracy

*Note rho will be different than the correlation slightly, as it is a slightly different test
Spearman Results between WPM and accuracy

Rho (correlation)	P-Value
0.52	1.92e-07

Table 13: Spearman Test Results between words per minute and accuracy %

Based on the results in Table 13 they are statistically significant. We will accept the alternative in favor of the null hypothesis that there is a correlation between typing speed and accuracy, which makes sense due to how the test is run.

The second largest correlation found is the relationship between words per minute and non a-z characters, doing a similar test as before with a significance level of 0.9

H0: There is no correlation between typing words per minute and non A-Z characters
HA: There is a correlation between typing words per minute and non A-Z characters

*Note rho will be different than the correlation slightly, as it is a slightly different test
Spearman Results between WPM and non A-Z Characters

Rho (correlation)	p-value
-0.22	0.036

Table 14: Spearman Test Results Between Words Per Minute and Non A-Z Characters

Again, the results are found to be statistically significant on Table 14. So the alternative hypothesis is accepted over the null hypothesis.

To be thorough, a correlation test was also run between typing accuracy and non-az characters.

H0: There is no correlation between typing accuracy and non A-Z characters
HA: There is a correlation between typing accuracy and non A-Z characters

*Note rho will be different than the correlation slightly as it is a slightly different test
Spearman Results Between Typing Accuracy and Non A-Z Characters

Rho (correlation)	p-value
-0.050	0642

Table 15: Spearman Test Results between Typing Accuracy and Non A-Z Characters

As expected, the results are not statistically significant with a significance level of 0.9 so the null hypothesis is favored over the alternative.

Finally, as a last consideration, two categorical variables were used. This includes passage type, and keyboard type. It would be good to consider what impact they have on WPM and accuracy, but do not want the differences between typists to impact the results. Thus, a linear mixed-effects model will be used. The STD between subjects was 0.50 WPM compared to a residual std of 7.16 STD. This again confirms that the difference between the two subjects is relatively small.

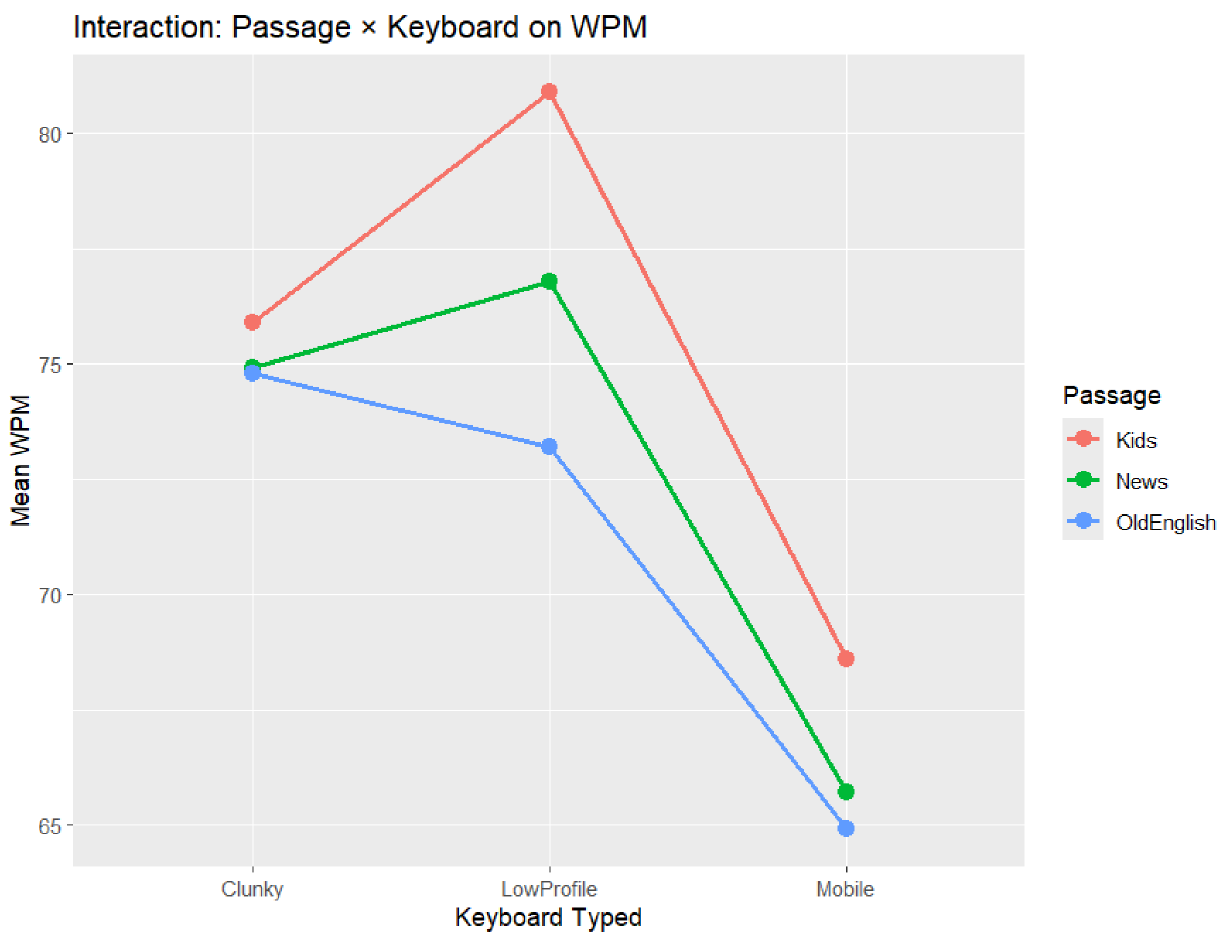

Figure 16: The Interaction between Passage and Keyboard Type on WPM

With a quick glance at Figure 16 there are two notable results. The most immediately obvious is that there seems to be a trend worth investigating that mobile keyboards have a lower WPM than both low-profile and clunky keyboards. Also, kids' passages seem to have an average WPM higher than news articles, which in turn has an average WPM higher than Old English articles.

*Note clunky keyboard and kids were regarded as the baseline

Predictor	Estimate	Std. Error	df	t	p
(Intercept)	75.90	2.29	37	33.12	< .001
Passage: News	-1.00	3.20	80	-0.31	.756
Passage: Old English	-1.10	3.20	80	-0.34	.732
Keyboard: Low Profile	5.00	3.20	80	1.56	.122
Keyboard: Mobile	-7.30	3.20	80	-2.28	.025
Passage × Low Profile (News)	-3.10	4.53	80	-0.68	.496
Passage × Low Profile (Old English)	-6.60	4.53	80	-1.46	.149
Passage × Mobile (News)	-1.90	4.53	80	-0.42	.676
Passage × Mobile (Old English)	-2.60	4.53	80	-0.57	.568

Figure 17: The Interaction between Passage and Keyboard Type on WPM

Looking through Figure 17, a few notable things come to mind. While not quite up to the statistical significance of 0.9, there is enough to raise eyebrows about low-profile keyboards potentially being correlated with a different typing speed than the baseline of a clunky keyboard, although no statistically significant difference was found here. Although there is a statistically significant difference in typing speed with the mobile keyboard, as seen in the chart that was seen. Surprisingly, there was no statistically significant correlation with the different passage types.

Now that there seems to be some difference between the keyboards, it would be beneficial to run a test specifically on the keyboards to see if there is a difference between them. For this, we found Estimated Marginal Means, which works for linear mixed models and can decide if there is a statistically significant difference between the types of keyboards, and the results are shown on Table 18.

Estimated Marginal Means for WPM Model by Keyboard

Comparision	Difference Estimate (WPM)	p-value
Clunky - LowProfile	-1.77	0.61
Clunky - Mobile	8.80	<.0001
Low Profile - Mobile	10.57	<0.001

Table 18: Estimated Marginal Means of Keyboards WPM

Table 18 also suggests what was found earlier. The statistically significant difference is between mobile keyboards and the other keyboards. While our test was only that they were different, based on the results, there is a good reason to suspect that mobile keyboards have a negative relationship to typing speed. This also matches the experience of both typists who thought it was more difficult to do the typing test with their phone.

A similar model was also run on the passage type, and the results were as follows.

Estimated Marginal Means for WPM Model by Passage

Comparision	Difference Estimate (WPM)	p-value
Kids - News	2.67	0.32
Kids - Old English	4.17	0.07
News - Old English	1.50	0.70

Table 19: Estimated Marginal Means of Passage Type WPM

As Table 19 shows, there is a statistically significant difference found between the Kids passages and Old English through the emmeans test.

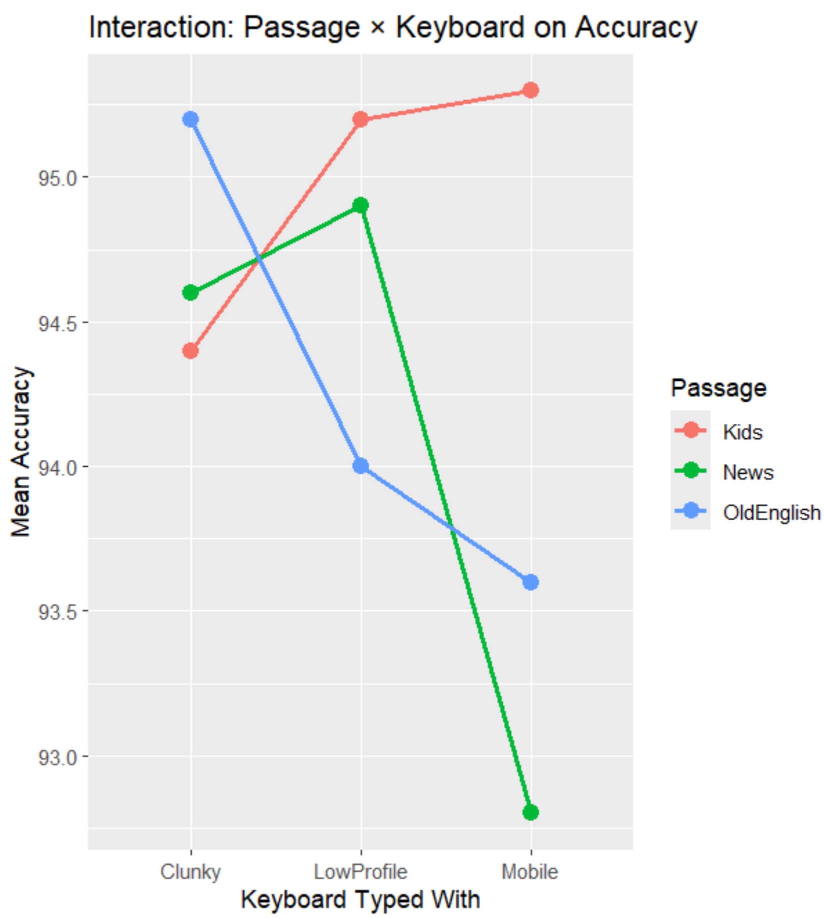

Figure 20: The Interaction between Passage and Keyboard Type on Accuracy (%)

A similar test was also run on the model for accuracy. Figure 20 shows how the keyboard type, passage, and mean accuracy interact.

The results are a bit all over the place in Figure 20. It is difficult to come up with a concrete order for the passages. It seems, however, that mobile keyboards still do have a lower accuracy than the other two from a quick glance which will be worth investigating. At the same time, accuracy in Old English and especially the news seems to take a pretty good hit on mobile keyboards.

The linear mixed effect model was taken on the accuracy, and the results are in Table 21 as follows.

Linear Mixed Effects Model for Typing Error Percent

Predictor	Estimate	Std. Error	df	t value	p-value
(Intercept)	94.40	0.66	14.20	142.13	< .001
Passage: News	0.20	0.86	80.00	0.23	.817
Passage: Old English	0.80	0.86	80.00	0.93	.356
Keyboard: Low Profile	0.80	0.86	80.00	0.93	.356
Keyboard: Mobile	0.90	0.86	80.00	1.04	.299
Passage × Low Profile (News)	–0.50	1.22	80.00	–0.41	.683
Passage × Low Profile (Old English)	–2.00	1.22	80.00	–1.64	.105
Passage × Mobile (News)	–2.70	1.22	80.00	–2.22	.030
Passage × Mobile (Old English)	–2.50	1.22	80.00	–2.05	.044

Table 21: The Interaction Between Passage and Keyboard Type on Accuracy

With a significance level of 0.10, not much seems to be statistically significant seen in Table 19 in terms of passage or keyboard type in terms of accuracy. Statistically significant results, however, do seem to be found with how mobile keyboards interact with both news passages (p val = 0.030) and Old English passages (0.044), which will be investigated further. For thoroughness, estimated marginal means with a t-test. While the results will not be discussed further as they are not significant, the results are in Tables 22 and 23 as follows.

Estimated Marginals Means for Accuracy by Keyboard

Comparision	Difference Estimate (Accuracy %)	p-value
Clunky - Low Profile	0.033	0.998
Clunky - Mobile	0.833	0.221
Low Profile - Mobile	0.800	0..248

Table 22: Estimated Marginal Means of Keyboards and Accuracy

Estimated Marginal Means for Accuracy by Passage Type

Comparision	Difference Estimate (Accuracy %)	p-value
Kids - News	0.867	0.196
Kids - Old English	0.700	0.342
News - Old English	-0.167	0.940

Table 23: Estimated Marginal Means of Passage Type and Accuracy

Going back to Table 21 it is notable how there seems to be a statistically significant difference in typing accuracy when combining typing on a mobile keyboard along with the more difficult article types, such as the News and Old English. A theory was that potentially this would be due to a higher prevalence of A-Z characters in the articles, which users are likely less accustomed to typing. To test this, the data was split, and as shown in Figure 24, kids' passages actually had the highest percentage of non-A-Z characters in the articles tested. Another theory is that due to the longer and more difficult spellings, mistakes were more likely to be made, although not tested for this project.

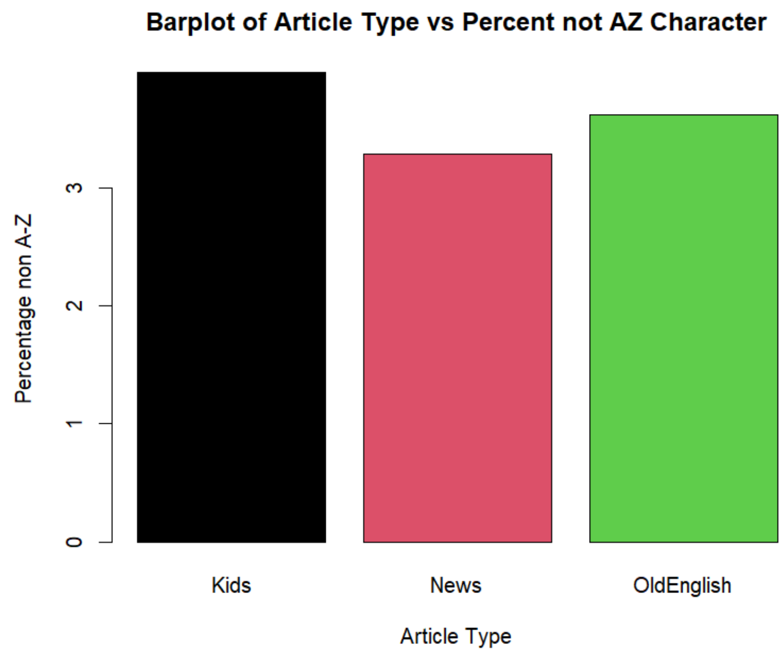

Figure 24: Barplot of Article Type vs Percent Non A-Z Characters

However, due to the design of the experiment, not all passages are equal and the effects of non A-Z characters may be more hidden than their surface level averages. By creating a model that accounts for the effects of non A-Z characters, it is clear they do have an impact on WPM.

Predictor	Estimate	Std. Error	t value
(Intercept)	84.7635	2.6901	31.509
PassageNews	-0.8008	2.7989	-0.286
PassageOldEnglish	-1.9963	2.8042	-0.712
KeyboardLowProfile	3.9045	2.8069	1.391
KeyboardMobile	-9.0428	2.8196	-3.207
NonAZChar	**-1.9918**	**0.3926**	**-5.074**
PassageNews:KeyboardLowProfile	-5.5898	3.9882	-1.402
PassageOldEnglish:KeyboardLowProfile	-5.5045	3.9637	-1.389

PassageNews:KeyboardMobile	-4.1408	3.9824	-1.040
PassageOldEnglish:KeyboardMobile	-3.1477	3.9593	-0.795

Table 25 The Interaction Between Passage, Keyboard Type, and Non A-Z characters on WPM

Table 25 doesn't show anything groundbreaking; however, it does clearly show, as would be intuitively assumed, that the percentage of non A-Z characters does have an impact. It reports that for each percentage increase in non A-Z characters, the subject's WPM is expected to drop by 2 words. Though this impact is particularly small, its impact is quite clear.

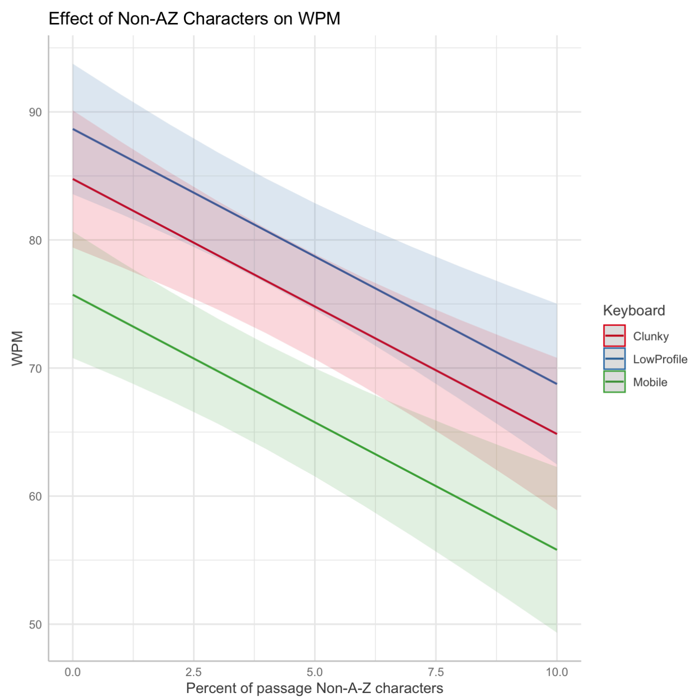

Figure 26 The effect of non A-Z characters on WPM for each keyboard type

This demonstrates that the impact of non A-Z characters, while only around 2 WPM per percent increase in the non A-Z character amount, can significantly reduce typing speeds when compounded.

Conclusion

Firstly, the mean typing speeds between the two subjects were fairly similar; however, there was a distinct difference observed between the distributions between two subjects. This would primarily indicate that there is much more individual variation between two subjects than is considered in the scope of this experiment. However, the differences were not distinct enough to invalidate the consideration of effects on both subjects. We primarily observed that there was no distinct difference between the low-profile and clunky keyboards. This is in sharp contrast to mobile keyboards, which significantly decreased the WPM for both subjects. Additionally, observed was the impact of Non-AZ characters, which had a constant impact across all keyboard types and decreased WPM as the percentage went up. This makes intuitive sense, as these characters require inputs outside the standard space for typing, and could even include several inputs at once. Additionally, it was shown there was a difference between kids books and Old English. While it may not necessarily be statistically significant at 5%, this result would make intuitive sense. However, far more research would be needed to indicate this difference, especially considering no difference was determined between two other groups which goes against the intuition of a difference existing.

Due to the observed variability in the distributions between the two subjects, this can make the conclusions from this study difficult to apply to a much wider population. Considering the population of this experiment, the results of these would only be relatively applicable to college students who are frequently using both a mobile device and a keyboard. Further, it is also important to consider the nature of this experiment. Both participants were instructed to sit down and warm up before these typing tests. Typing tests are also not entirely representative of a standard typing process that often includes many revisions, pauses, and with the content being typed already considered mentally. Though general trends could be important when considering the upper bounds of typing speeds. However it is important to further consider that strong individual biases likely exist, with several other factors that could be considered, social media screentime, gaming habits (including what keyboard type is used?), general typing habits (often typing up long reports?), mood/sleep, and the typing environment. Further studies on a far broader range of individuals could consider some or all of these factors that could and likely do impact typing speed across individuals.